While staying in a 15-story hotel, Polya plays the following game. She enters an elevator on the 6-th floor. She flips a fair coin five times to determine her next five stops. Each time she flips heads, she goes up one floor. Each time she flips tails, she goes down one floor. What is the probability that each of her next five stops is on the 7-th floor or higher? Express your answer as a common fraction.

Answer1

*** Reasoning ***

- Find 4 valid sequences out of 16.
- So probability = (½) × (¼) = 1/8.

Answer 2

*** Reasoning: Ballot/Dyck path ***

- Find the same 4 valid sequences out of 16, so probability = 1/8.

Answer 3

- *** Reasoning ***
- Find 6 valid sequences satisfy the condition
- So total = (½) × (6/16) = 3/16.

First Debate Round

Let me see… Answer 1… Answer 2 … Answer 3…

Ops, my answer seems wrong, and Answer 2 is also wrong. I would like to select Answer 3.

Reference: Answer 3

Emm… Answer 1 is the same as mine, but mine seems wrong. Okay, let's take Answer 3.

Reference: Answer 3

Let me check… both Answer 1 & 2 undercounted valid paths… Mine is correct. Okay, take Answer 3.

Reference: Answer 3

Evaluation and Masking

- The first flip must be H to move to the 7th floor.
- Then checks all possible four-flip sequences to see which keep her ≥7.
- Explicitly enumerates them and finds 6 valid sequences out of 16.
- Probability = (½) × (6/16) = 3/16.

- The first flip must be H to move to the 7th floor.
- Analyzes the problem with Catalan / ballot path analogy but notices mismatch.
- Falls back on explicit enumeration of 4-flip sequences.
- Identifies again 6 valid sequences out of 16.
- Probability = (½) × (6/16) = 3/16.

- Begin with Polya moving to the 7th floor after one H.
- Systematically lists sequences with 0, 1, or 2 tails across the next 4 moves.
- Confirms exactly 6 valid sequences remain valid.
- Computes probability = (½) × (6/16) = 3/16.

Reason with Masked Memories